

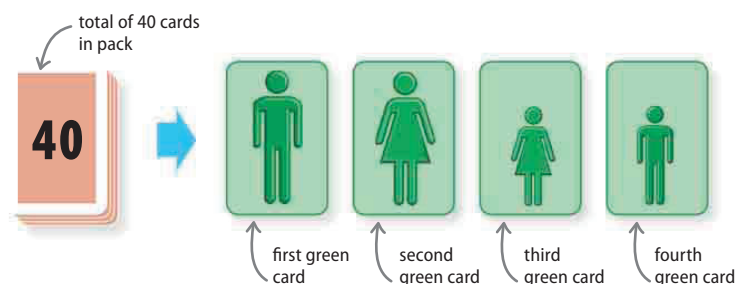


Dependent events

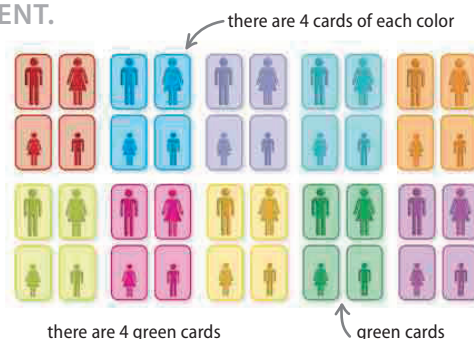
THE CHANCES OF SOMETHING HAPPENING CAN CHANGE ACCORDING TO THE EVENTS THAT PRECEDED IT. THIS IS A DEPENDENT EVENT.

Dependent events

In this example, the probability of picking any one of four green cards from a pack of 40 is 4 out of 40 (4/40). It is an independent event. However, the probability of the second card picked being green depends on the color of the card picked first. This is known as a dependent event.



▷ **Color-coded**
This pack of cards contains 10 groups, each with its own color. There are 4 cards in each group.



$$\frac{4}{40}$$

◁ What are the chances?

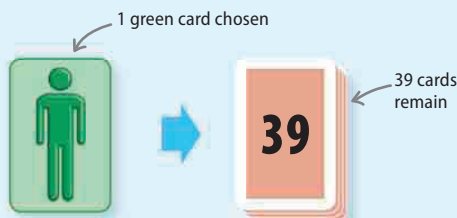
The chances of the first card picked being green is 4 in 40 (4/40). This is independent of other events because it is the first event.

there are 40 cards in total

Dependent events and decreasing probability

If the first card chosen from a pack of 40 is one of the 4 green cards, then the chances that the next card is green are reduced to 3 in 39 (3/39). This example shows how the chances of a green card being picked next gradually shrink to zero.

The first card is chosen and is green. The next card picked will be chosen from the remaining 39 cards.



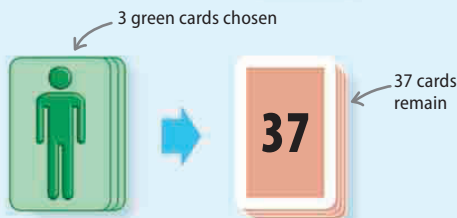
$$\frac{3}{39}$$

chances of next card being green

cards left

◁ **The chances of the next card** picked being green are 3 in 39 (3/39). With 1 of the 4 green cards picked there are 3 left in the pack.

The first 3 cards have been chosen and were all green. The next card to be picked will be chosen from the remaining 37 cards.



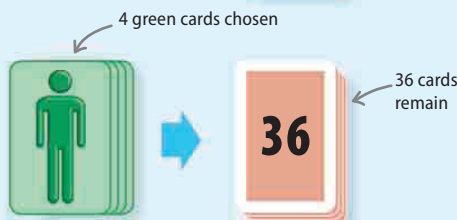
$$\frac{1}{37}$$

chances of next card being green

cards left

◁ **The chances of the next card** being green are 1 in 37 (1/37). With 3 of the 4 green cards picked there is 1 green card left in the pack.

The first 4 cards have been chosen and were all green. The next card to be chosen will be picked from the remaining 36 cards.



$$\frac{0}{36} = 0$$

chances of next card being green as all green cards are picked

cards left

◁ **The chances of the next card** being green are 0 in 36, or zero. With all 4 green cards picked there are no green cards left in the pack.